Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **control and automation software**, **state machines**, **machine vision**, and **automated test systems**, plus closed-loop autonomous-driving research. Works directly with customers.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built a fault-tolerant **job orchestration service** with **explicit state transitions**, step-level failure isolation, and automated recovery, processing 1K+ jobs and tripling throughput under concurrent load without cascading failures.
- Implemented **validation and reconciliation checks** with CI/CD-backed releases across 5K+ monthly integration jobs, maintaining **99%+ production reliability** while cutting data-integration latency by 35% through query tuning and indexing.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Integrated **computer vision** and **OCR** (GPT-4o Vision, Qwen-VL) into an automated grading pipeline with a Claude Vision fallback path, evaluating 10K+ handwritten and text submissions daily with 90%+ grading consistency.
- Designed a **concurrent batch-processing engine** using multithreading with fault-tolerant execution and per-call cost tracking, processing 1K+ files per batch and cutting feedback turnaround by 60% for 150+ instructors.

USC AD Labs **Los Angeles, CA**
Research Assistant - Autonomous Driving *January 2025 – May 2025*

- Built a GPT-based Decision Transformer for offline **reinforcement learning** in **autonomous driving**, improving long-horizon trajectory planning by 18% in simulation and building data pipelines that process 100K+ trajectory logs.
- Designed a map-aware spatial-temporal encoder fusing HD-map geometry with ego-vehicle trajectory history via multi-head attention, improving **closed-loop planning stability** by 15% with fewer off-road trajectories across simulation rollouts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Automated nightly **build, install, and test** cycles on dedicated VMs with Jenkins, running 20+ integration and unit test suites across Windows, Linux, and macOS and generating failure reports for rapid triage.
- Troubleshot and optimized large-scale **API integrations** using OOP design patterns, AWS Glue, and S3 in Agile sprints with Microsoft/GitHub stakeholders, improving response times 30% and pipeline efficiency 68%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Gathered requirements and **presented the system architecture to the client**, mapping 12+ integration points across mobile apps, Flask APIs, Firebase, and payment webhooks before implementation, reducing design errors by 25%.
- Built Flask APIs with Firebase **real-time synchronization** and transaction locking to prevent double-booking of shared time slots, sustaining sub-200ms booking latency across 50K+ records on Android, iOS, and web.

Skills

Controls & Automation: PC-Based Control Software, PLC-Style Scan-Cycle Logic, PackML State Machines, Motion Control (Homing, Trapezoidal Profiles), Safety Interlocks, 4-20 mA Analog I/O, Modbus TCP, Fault Diagnostics
Machine Vision & ML: OpenCV, Machine Vision Inspection, PyTorch, Transformers, OCR
Languages: Python, C++, Java, JavaScript/TypeScript, SQL, Bash
Tools & Platforms: Linux, Git, PyTest, Jenkins, GitHub Actions, Docker, AWS, Azure
Engineering Practices: Test Automation, Integration Testing, Simulation, Documentation, Root-Cause Analysis, Agile

Projects

AutoCell | *Python, OpenCV, Modbus TCP, pytest*

- Built **PC-based control software** for a simulated pick-and-place inspection cell: 5 ms **PLC-style scan cycle**, PackML **state machine**, E-stop/guard/light-curtain **interlocks**, and servo axes with homing, trapezoidal profiles, and following-error monitoring.
- Added **OpenCV machine vision** inspection, 4-20 mA analog diagnostics, per-part **traceability**, a **Modbus TCP** HMI server, and an **I/O checkout**, verified by 31 tests covering injected faults and recovery paths.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** in C++ with Gantt-chart visualization, validating correctness across 1K+ simulated process combinations and cutting debugging time by 20%.